

* Reinforce Algorithm :-

1. Policy (π_θ):

A parameterized function that defines the probability of taking an action given a state. The parameters of the policy are denoted by θ .

2. Trajectories (τ):

A sequence of states, actions and rewards experienced by the agent in an episode.
 $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_T, a_T, r_T)$.

3. Return (G_t):

The cumulative discounted reward from time step t .

$$G_t = \sum_{k=t}^T \gamma^{k-t} r_k$$

where,

γ is the discount factor.

4. Objective :

Maximize the expected return

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} (G_0).$$

Q. Consider a simple grid world with 3 states: $S = \{s_0, s_1, s_2\}$. The agent starts in s_0 and can take two actions $A = \{a_0, a_1\}$. The episode ends when the agent reaches s_2 .

- The rewards are as follows:

$$r(s_0, a_0) = 1, \quad r(s_0, a_1) = 0$$

$$r(s_1, a_0) = 2, \quad r(s_1, a_1) = -1$$

$$r(s_2, \cdot) = 0 \text{ (terminal state)}$$

- The policy π_0 is parameterized by θ , and the action probabilities are:

$$\pi_0(a_0|s_0) = 0.6, \quad \pi_0(a_1|s_0) = 0.4$$

$$\pi_0(a_0|s_1) = 0.7, \quad \pi_0(a_1|s_1) = 0.3$$

- The discount factor $\gamma = 0.9$

- i. Compute the returns G_t for each time step in the trajectory.
- ii. Compute the log probabilities of the action taken under the policy π_0 .
- iii. Compute the gradient of the policy parameters θ using the Reinforce Algo.
- iv. Update the policy parameters θ using gradient ascent with a learning rate $\alpha = 0.1$.

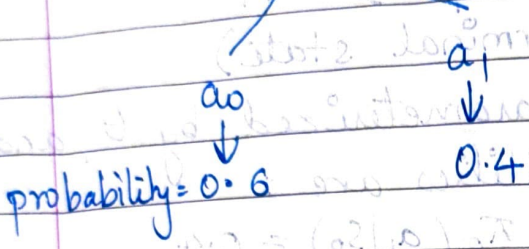
OR

- i. Update the policy parameter θ using gradient ascent using $\alpha = 0.1$, $\gamma = 0.9$ and $\tau = (s_0, a_0, r_0 = 1, s_1, a_0, r_1 = 2, s_2)$

Solution :-

→ Episode Trajectory :-

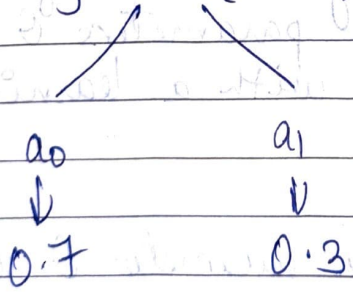
(1) ~~At~~ start at S_0 :
 policy $\pi_0(a|S_0)$



Agent samples a_0
 ∴ the environment transit to S_1 & provides a reward $r_0 = 1$

Record $(S_0, a_0, r_0 = 1, S_1)$

Start at S_1
 policy $\pi_0(a|S_1)$



Agent samples a_0
 ∴ the env. transit to S_2 & provides a reward $r_1 = 2$

Record $(S_1, a_0, r_1 = 2, S_2)$

∴ Agent follows the policy and generates the foll. trajectory :

$$z = (s_0, a_0, r_0 = 1, s_1, a_0, r_1 = 2, s_2)$$

→ Compute Returns :-

The return G_t is the cumulative discounted reward from time step t onwards. For this trajectory :

At $t=0$ (state s_0) :

$$\begin{aligned} G_0 &= r_0 + \gamma r_1 + \gamma^2 r_2 \\ &= 1 + 0.9 \times 2 + (0.9)^2 \times 0 \\ &= 2.8 \end{aligned}$$

At $t=1$ (state s_1) :

$$\begin{aligned} G_1 &= r_1 + \gamma r_2 \\ &= 2 + 0.9 \times 0 \\ &= 2 \end{aligned}$$

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$$

→ Compute log Probabilities :-

The log probability of the actions taken under the policy π_θ :

At $t=0$ (state s_0 , action a_0) :

$$\begin{aligned} \log \pi_\theta(a_0 | s_0) &= \log(0.6) \\ &\approx -0.5108 \end{aligned}$$

At $t=1$ (state s_1 , action a_0) :

$$\begin{aligned} \log \pi_\theta(a_0 | s_1) &= \log(0.7) \\ &\approx -0.3567 \end{aligned}$$

→ Compute Gradient :

The gradient of the objective $J(\theta)$ is :

$$\nabla_{\theta} J(\theta) = \sum_{t=0} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t$$

For this episode :

At $t=0$:

$$\begin{aligned} & \nabla_{\theta} \log \pi_{\theta}(a_0 | s_0) \cdot G_0 \\ &= -0.5108 \times 2.8 \\ &\approx -1.4302 \end{aligned}$$

At $t=1$

$$\begin{aligned} & \nabla_{\theta} \log \pi_{\theta}(a_0 | s_1) \cdot G_1 \\ &= -0.3567 \times 2 \\ &\approx -0.7134 \end{aligned}$$

∴ The total gradient for this episode is :

$$\begin{aligned} \nabla_{\theta} J(\theta) &\approx -1.4302 + (-0.7134) \\ &\approx -2.1436 \end{aligned}$$

→ Update Policy Parameters :

Assume $\alpha = 0.1$. The policy parameter θ are updated as :

$$\begin{aligned} \theta_{\text{new}} &= \theta_{\text{old}} + \alpha \cdot \nabla_{\theta} J(\theta) \\ &= \theta_{\text{old}} + 0.1 \times (-2.1436) \\ &= \theta_{\text{old}} - 0.2144 \end{aligned}$$

Assuming θ_{old} to be 1.0

$$\therefore \theta_{new} = \theta_{old} - 0.2144$$

$$= 1.0 - 0.2144$$

$$\therefore \theta_{new} = 0.7856$$

→ Interpretation

- The policy parameter θ is updated from 1.0 to 0.7856.
- This update reduces the value of θ , which means the policy is adjusted to decrease the likelihood of the actions taken in this episode (since the gradient was -ve).
- Over multiple episodes, the policy will continue to adjust to maximize the expected return, i.e., the model is placing more weight on future rewards to maximize long-term results.